



Laptop price

Analysis





welcome to the

PRESENTATION

By :

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- **Panshu Jain =72519240**
 - **Karmanpreet Singh=72514059**
 - **Harkirat Singh=72514058**

Machine Learning is when a computer learns from data to make predictions without being explicitly programmed.

Types of machine learning:

- **Supervised ML : Learns from the labelled data.**
- **. Unsupervised ML : Learns from the unlabelled data**
- **Reinforcement ML. : Learns by receiving rewards and penalties**

*Machine
Learning*



introduction to the project



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- **Laptop prices vary based on specifications and brand**
 - **Data analysis helps understand pricing patterns**
 - **Python is used for visualization and insights.**
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DATA DESCRIPTION



df.head()

	brand	model	screen_size	color	harddisk	cpu	ram	OS	special_features	graphics	graphics_coprocessor	cpu_speed	rating	Price	Sale Product Count	Total Sales	Average
0	ROKC	NaN	14 Inches	Blue	1000 GB	Intel Core i7	8 GB	Windows 11	NaN	Integrated	Intel	1.2 GHz	NaN	\$1,783.99	14	24975.86	
1	HP	NaN	15.6 Inches	Silver	1000 GB	Intel Core i5	64 GB	Windows 11 Pro	Backlit Keyboard	Integrated	Intel	NaN	4.5	\$449.00	60	26940	
2	MSI	Vector GP66 12UGS-267	15.66 Inches	Core Black	NaN	Intel Core i9	32 GB	Windows 11 Home	NaN	Dedicated	NaN	1.8 GHz	5.0	\$999.99	24	23999.76	
3	Apple	MacBook Air	13.3 Inches	Silver	256 GB	Unknown	8 GB	Mac OS	Backlit Keyboard	Integrated	NaN	NaN	4.8	\$3,423.99	60	205439.4	
4	Apple	MacBook Air	15.3 Inches	Midnight	256 GB	Unknown	8 GB	Mac OS	NaN	Integrated	NaN	NaN	4.8	\$699.00	33	23067	





Portfolio by Juliana Silva



- Histogram
- Boxplot
- Bar graph
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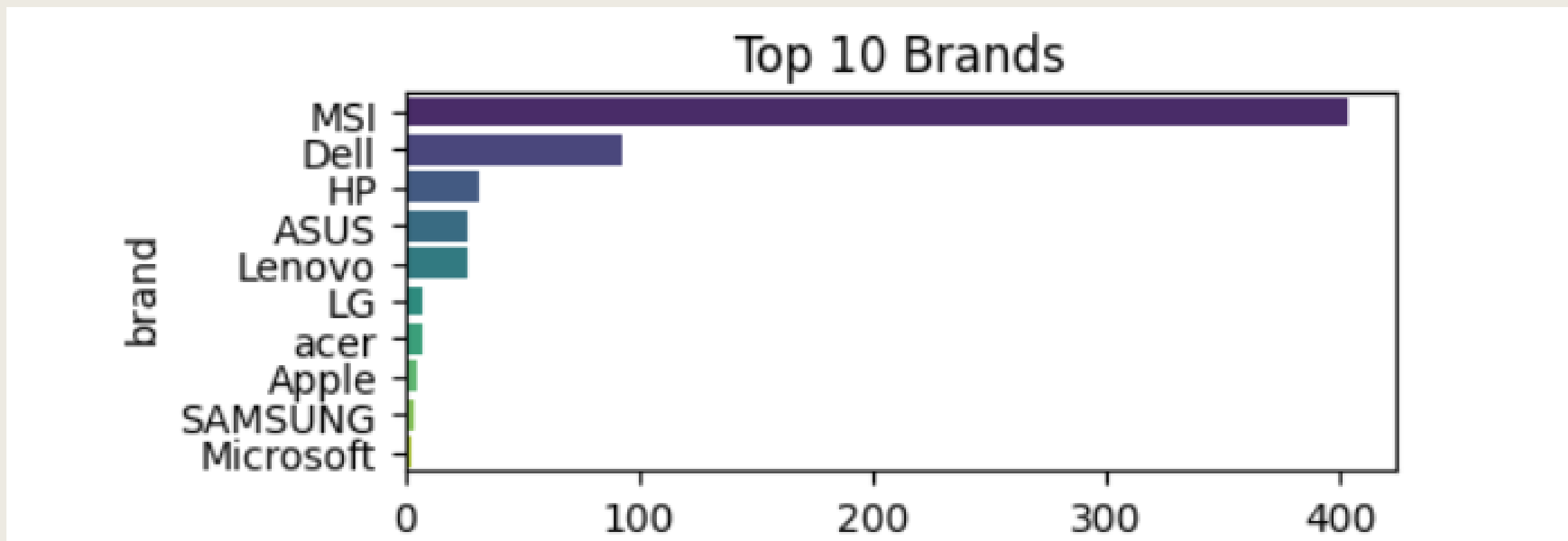
VISUALIZATION INSIGHTS



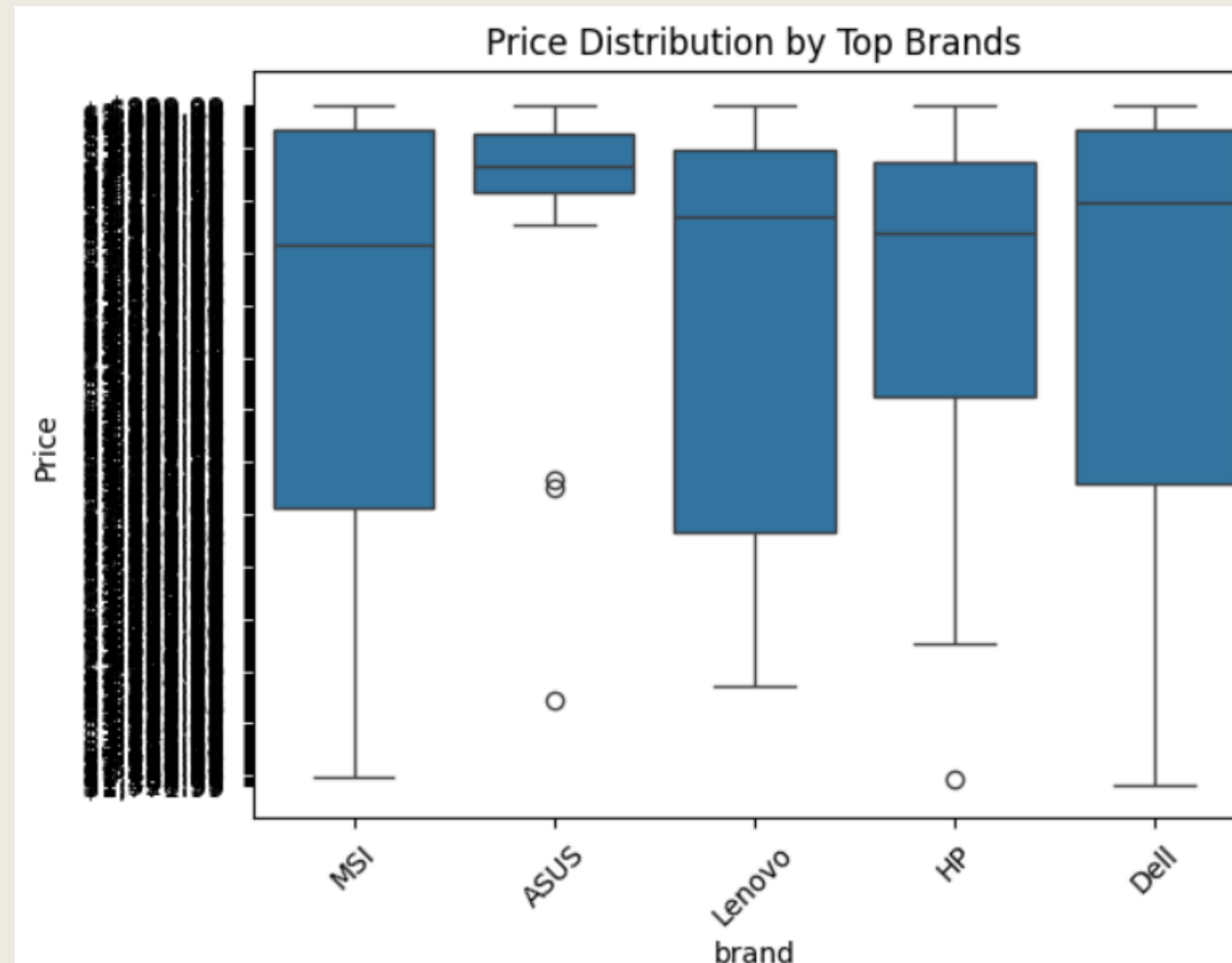
HISTOGRAM



A histogram is a graph that shows how data is distributed by grouping values into ranges (bins).



BOXPLOT



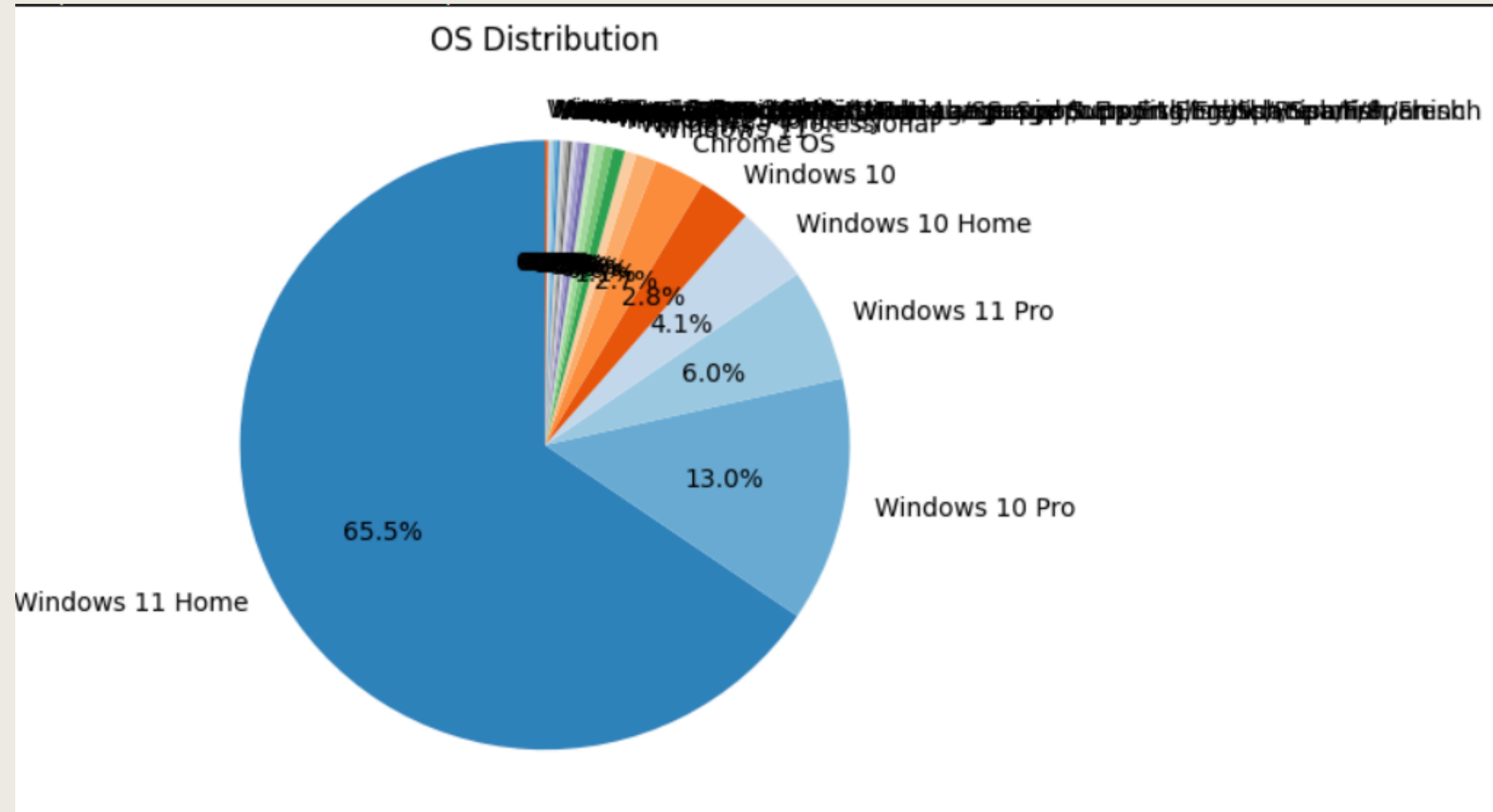
The chart shows the price distribution of laptops across top brands (MSI, ASUS, Lenovo, HP, Dell). Each box represents the range and spread of prices, with the middle line indicating the median price.

ASUS and Dell tend to have higher median prices, suggesting more premium offerings.

HP and Lenovo show wider variability, meaning they offer both budget and high-end options.

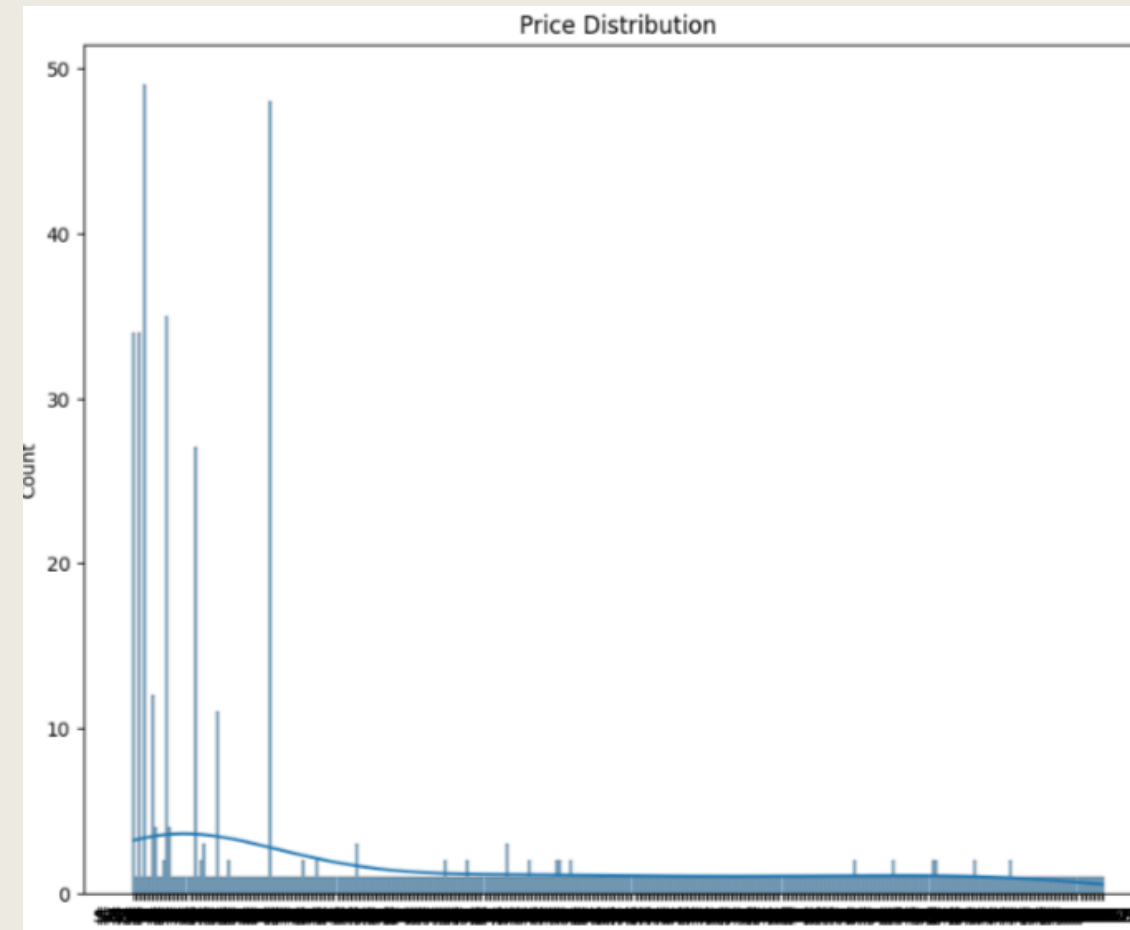
Outliers indicate extremely high or low-priced models compared to the typical range within each brand.

Pie chart



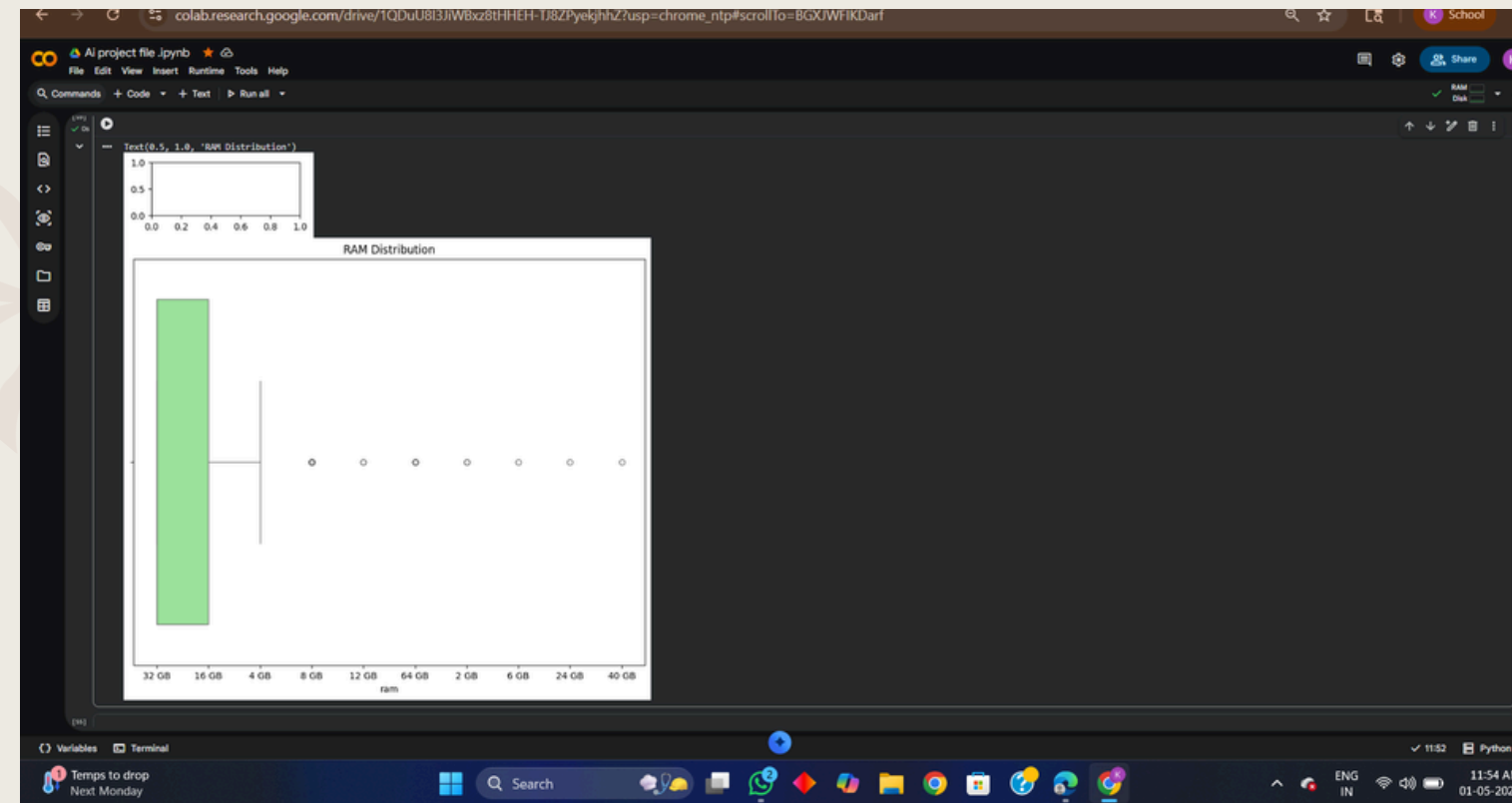
- This chart shows which operating systems are used in laptops.
- Windows 11 Home is the most common and used in most laptops.
- Windows 10 Pro and Windows 11 Pro are used less but still noticeable.
- Other systems like Windows 10 Home and Chrome OS are used by fewer people.
- Some operating systems are very rare and used in very few laptops.

Line Plot

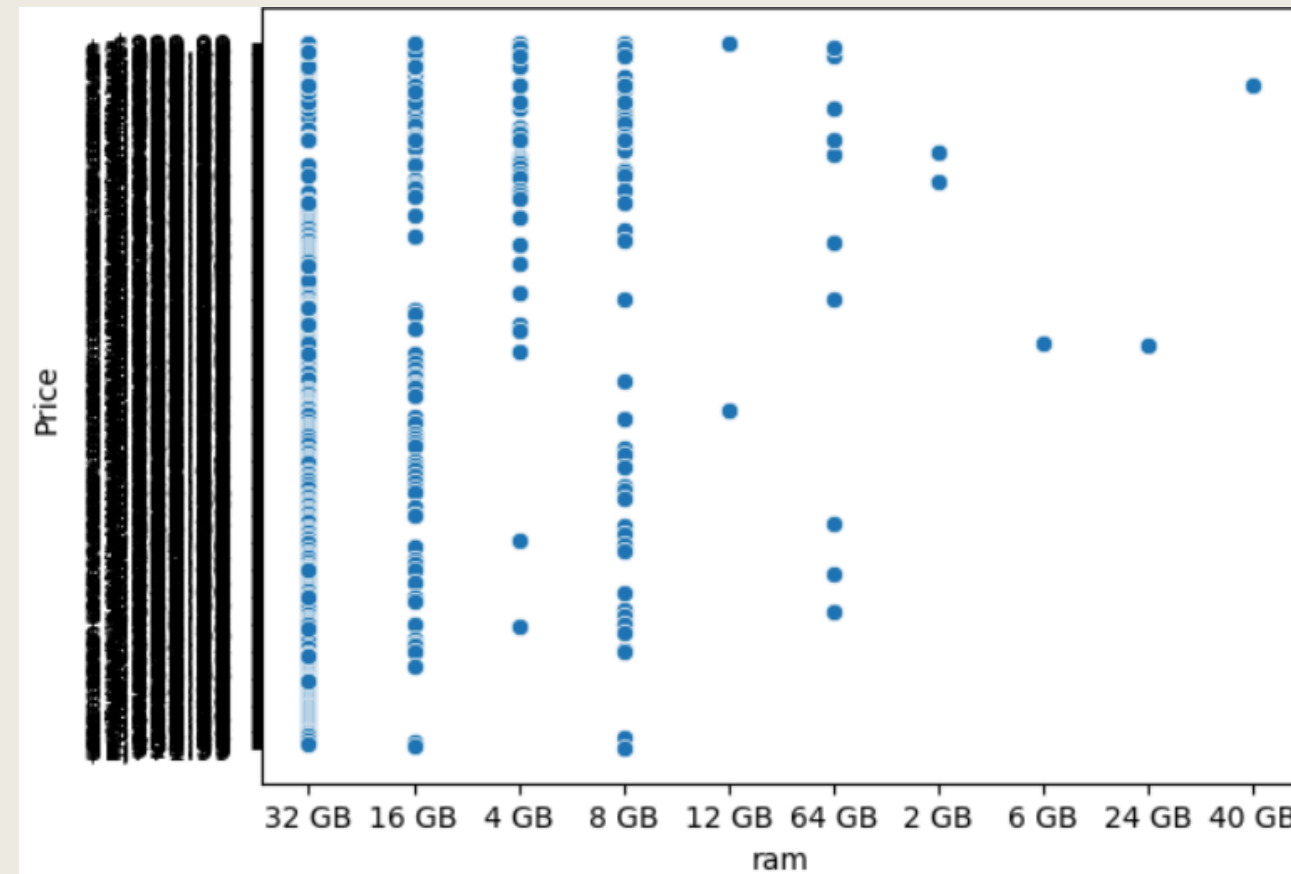


- **This chart shows how much laptops usually cost.**
- **Most laptops are not very expensive.**
- **Only a few laptops are very costly.**
- **As price goes up, the number of laptops goes down.**
- **So, cheap and mid-range laptops are more common.**

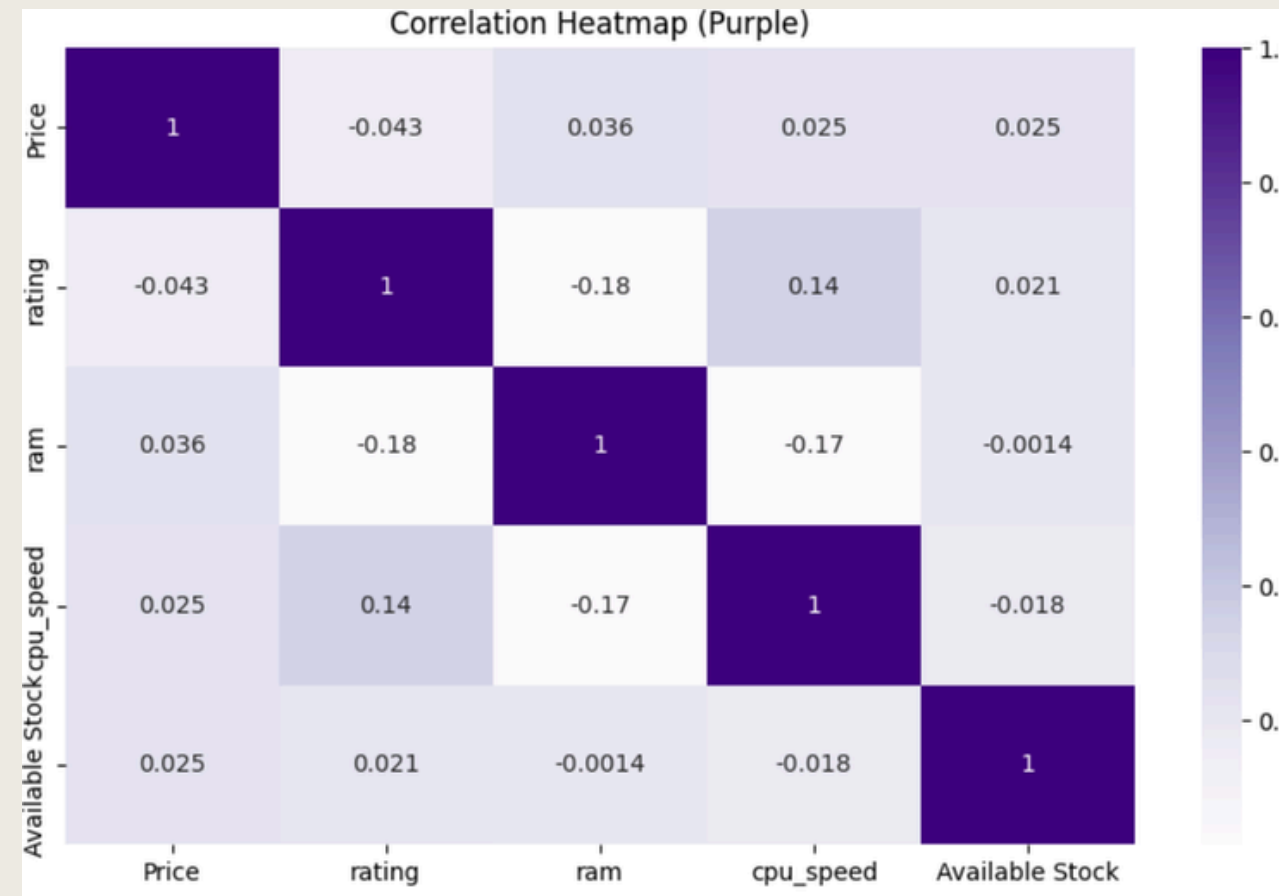
Box plot chart



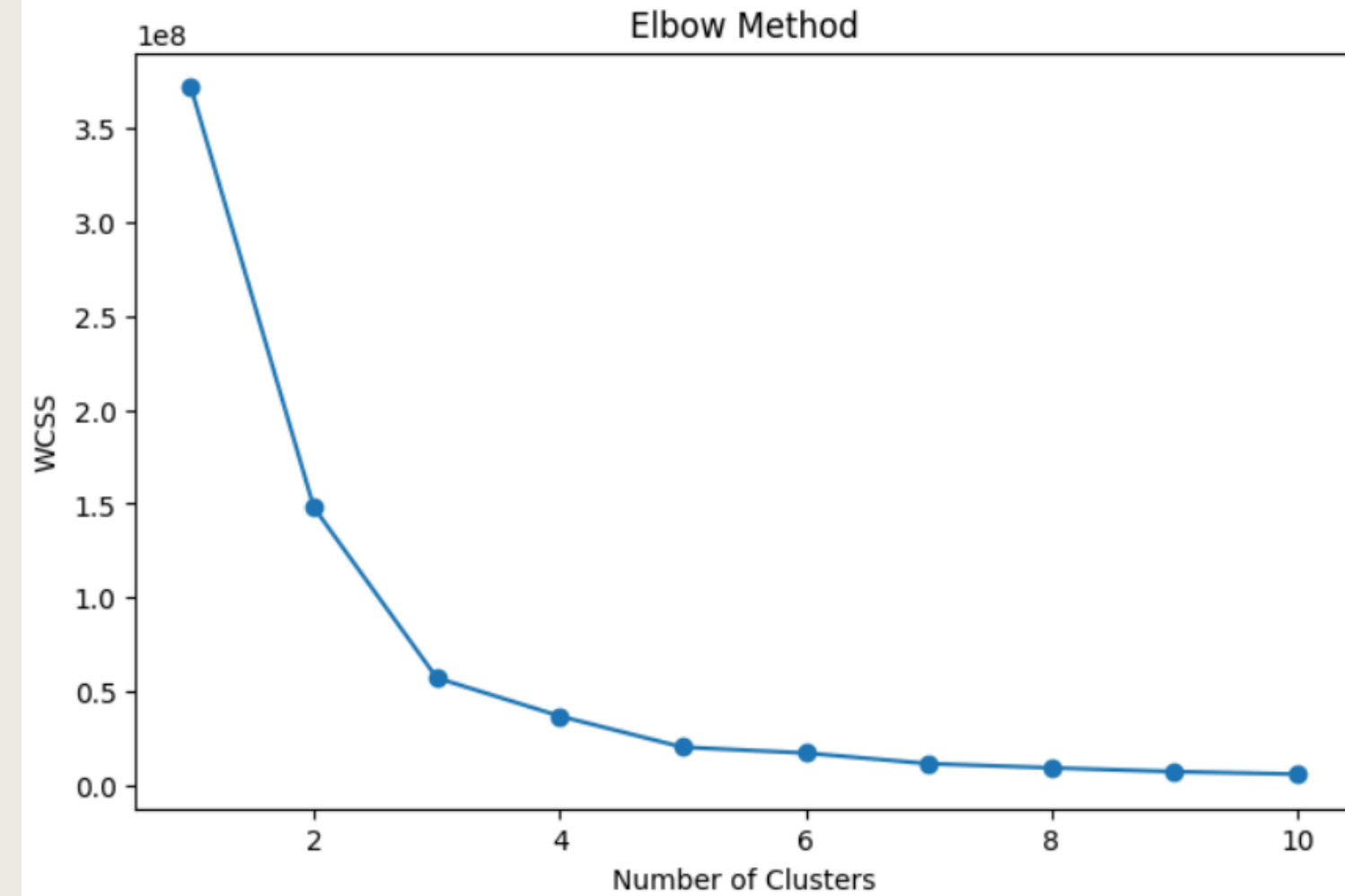
- This chart shows how much RAM laptops usually have.
- Most laptops come with 8 GB to 16 GB RAM.
- A few laptops have very high RAM (like 32 GB or more).
- Some laptops have low RAM (like 2 GB or 4 GB), but they are fewer.
- So, medium RAM laptops are the most common.



- THIS CHART SHOWS THE RELATIONSHIP BETWEEN RAM AND LAPTOP PRICE.
- LAPTOPS WITH MORE RAM USUALLY COST MORE.
- MOST LAPTOPS ARE IN THE 8 GB TO 16 GB RANGE WITH DIFFERENT PRICES.
- SOME HIGH RAM LAPTOPS (LIKE 32 GB OR MORE) ARE MORE EXPENSIVE.
- OVERALL, PRICE INCREASES AS RAM INCREASES, BUT NOT ALWAYS EXACTLY THE SAME.



- **THIS CHART SHOWS THE RELATIONSHIP BETWEEN DIFFERENT LAPTOP FEATURES.**
- **THE VALUES RANGE FROM -1 TO 1, WHERE VALUES CLOSER TO 1 MEAN STRONG RELATION.**
- **MOST VALUES ARE CLOSE TO 0, MEANING THERE IS NO STRONG RELATIONSHIP.**
- **PRICE HAS VERY WEAK CONNECTION WITH RAM, RATING, AND CPU SPEED.**
- **OVERALL, THE FEATURES ARE MOSTLY INDEPENDENT AND DON'T STRONGLY AFFECT EACH OTHER.**



- **THIS GRAPH SHOWS HOW THE NUMBER OF CLUSTERS AFFECTS THE MODEL PERFORMANCE (WCSS VALUE).**
- **AS THE NUMBER OF CLUSTERS INCREASES, THE WCSS VALUE DECREASES QUICKLY AT FIRST.**
- **AFTER AROUND 3–4 CLUSTERS, THE DECREASE BECOMES SLOWER AND MORE STABLE.**
- **THE “ELBOW POINT” (BEND IN THE GRAPH) APPEARS NEAR $K = 3$ OR 4.**
- **THIS MEANS 3–4 CLUSTERS IS THE BEST CHOICE FOR GROUPING THE DATA EFFICIENTLY.**



Portfolio by Juliana Silva



thanks
FOR WATCHING



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